

SIOPTR Course

Pós-Graduação em Sistemas Computacionais Embebidos
Sistemas Operativos de Tempo-Real (PGSCE-SIOPTR)

2025/26

Paulo Baltarejo Sousa and Ricardo Severino

`{pbs,rar}@isep.ipp.pt`



Instituto Superior de
Engenharia do Porto

P.PORTO

Material and Slides

Some of the material/slides are adapted from various:

- Presentations found on the internet;
- Books;
- Web sites;
- ...

Outline

- 1 Overview**
- 2 Assessment**
- 3 Planning**
- 4 Instructors**

- 1 Overview**
- 2 Assessment
- 3 Planning
- 4 Instructors

Course

- Sistemas Operativos de Tempo-Real (PGSCE-SIOPTR)
- Contact hours: 30
- ECTS: 3
- This course begins by addressing the essential concepts and principles of real-time (RT) systems, such as scheduling algorithms and resource-sharing mechanisms.
- It also presents the basic concepts and principles of real-time operating systems (OSs), together with the mechanisms used to ensure the correct operation of applications with real-time requirements.
- A course on advanced OS concepts is crucial for computer science education in critical computational systems.

Overview

- This course focuses on **operating system topics**, with special focus on Real-time.
- It is assumed that students are familiar with:
 - Computer architecture concepts
 - Operating systems usage and management
 - Data structures
 - Advanced C programming
 - Basic Linux knowledge

Goals

- At the end of the course, the students should be able to:
 - **Understand** concepts and principles associated with real-time systems
 - **Evaluate** scheduling algorithms and schedulability analysis for uni- and multi-processor systems;
 - **Evaluate** mechanisms for shared resources management in real-time systems;
 - **Understand** fundamental RTOS concepts and principles as well as architecture;
 - **Evaluate** OS implemented mechanisms to support real-time requirements;
 - **Apply** RTOS mechanisms to support real-time constrained applications.

Topics

- The course delivers the following main topics:
 - **Fundamental Real-Time** principles, concepts, and algorithms;
 - **Fundamental Operating Systems** principles, components, and architectures;
 - **Scheduling** and **shared resources**;
 - **Kernel development**.

Teaching/learning methodology

- Lectures (TP)
 - Expose students to various concepts of Real-Time Systems and also Operating Systems with special focus on Linux kernel.
- Labs (PL)
 - Students will solve a set of simple exercises.
 - Students will be encouraged to solve more complex exercises.

Teaching Material (I)

- Support material available at the subject's Moodle page.
 - **TP pdf files are for supporting TP classes.**
 - **PL pdf files are for supporting PL classes.**
 - **TT pdf files are for students' self study.**

Teaching Material (II)

- Books:
 - Giorgio C. Buttazzo (2011). "Hard Real-Time Computing Systems: Predictable Scheduling Algorithms and Applications (3rd ed.)". Springer Publishing Company.
 - Bovet and Cesati (2010). Understanding the Linux Kernel, 3rd edition. O'Reilly Media, Inc.
 - Robert Love (2010). Linux Kernel Development, 3rd edition. Addison-Wesley Professional.
 - Jim Cooling (2019). Real-time Operating Systems Book 1: The Theory.
 - Jim Cooling (2017). Real-time Operating Systems Book 2 - The Practice: Using STM Cube, FreeRTOS and the STM32 Discovery Board (Engineering of Real-Time Embedded Systems).

Teaching Material (III)

- Surveys:
 - Robert I. Davis and Alan Burns. "A survey of hard real-time scheduling for multiprocessor systems". ACM Comput. Surv. 43, 4, Article 35 (October 2011).
 - Alan Burns and Robert I. Davis. "A Survey of Research into Mixed Criticality Systems". ACM Comput. Surv. 50, 6, Article 82 (November 2017).
- Online resources:
 - <http://www.kernel.org>
 - <https://elixir.bootlin.com/linux/latest/source>
 - <http://www.linux.org/>
 - <https://lkml.org/>
 - <http://lwn.net/>
- Tools:
 - Virtual Box

- 1 Overview
- 2 Assessment**
- 3 Planning
- 4 Instructors

Moments & Grade

- Assessment will have two **Moments**:
 - Moment 1 (M1): Real-time Lab exercise;
 - Moment 2 (M2): Operating System Lab exercise.
- These Moments (M1 and M2) are:
 - **Mandatory for all students, regardless of their status.**
 - M1, M2, and E are graded in the interval [0.00,20.00];
- **The final grade (CF) is determined as follows:**
 - $CF = M1 * 0.40 + M2 * 0.60$.
 - All CF components are graded in the interval [0.00,20.00]

- 1 Overview
- 2 Assessment
- 3 Planning**
- 4 Instructors

Week mapping ¹

	Tue. (online)	Thu. (online)	Sat. (on site)
Week 1	21/04	23/04	–
Week 2	28/04	30/04	02/05
Week 3	05/05	07/05	09/05 (M1)
Week 4	12/05	14/05	16/05
Week 5	19/05	21/05	23/05
Week 6	–	–	30/05 (M2)

- M1: 09/05/2026
- M2: 30/05/2026

¹We consider the first week day is on Monday and the last week day is on Sunday.

- 1 Overview
- 2 Assessment
- 3 Planning
- 4 Instructors**

Identification

- Paulo Baltarejo Sousa.
 - E-Mail: `pbs@isep.ipp.pt`.
- Ricardo Severino.
 - E-Mail: `rar@isep.ipp.pt`.

Support

- There is no specific schedule for students support.
 - Whenever students need support, must send an email to any instructor.

Email rule

- **Subject/Assunto: SIOPTR**
 - If you do not follow, the email is ignored (i.e., it is not "received")